DAY 04 NOTES

Supervised Learning, Classification & Regression

Beginner-friendly theory notes • Concepts covered in class

Learning Goals

- Understand supervised learning and why it uses labeled data.
- Identify features and labels in a supervised learning problem.
- Understand the training-to-prediction workflow.
- Distinguish classification from regression.
- Explain binary, multiclass, and multilabel classification.

Big picture: Supervised Learning → Labeled Training Data → Training → Trained Model → New Input → Prediction

1. What Is Supervised Learning?

Supervised learning is a type of Machine Learning in which a model learns patterns from labeled training data and uses those patterns to predict answers for new data.

Think of a teacher showing a student examples with correct answers. The student learns from those examples and then tries a new question. In supervised learning, the labeled examples play a similar role.

Key idea: Training examples contain known target answers. The model learns patterns that connect input information to those answers.

Example: Student result prediction

- Student 1: Study hours = 5; attendance = 90%; result = Pass.
- Student 2: Study hours = 2; attendance = 50%; result = Fail.
- Student 3: Study hours = 6; attendance = 95%; result = Pass.

If the goal is to predict the result, study hours and attendance are **features**, while Pass or Fail is the **label**. These features may help make a prediction, but they do not guarantee the result.

Features and labels

Features: Input information used by a model to make a prediction. Examples include house area, customer income, email text, or visual information from an image.

Label: The known target answer the model is trained to predict. Labels can be categories (Pass/Fail) or numbers (a house price).

Important: A column is not automatically a feature or a label. Its role depends on what the model is asked to predict.

2. How Supervised Learning Works

Step 1 — Collect historical data

Gather examples related to the problem. For house-price prediction, records might include area, bedrooms, location, and actual selling price.

Step 2 — Identify features and label

For this task, area, bedrooms, and location are features. The actual selling price is the label.

Step 3 — Train the model

A Machine Learning algorithm processes labeled examples and learns patterns connecting features to labels. The learned system is called the trained model.

Step 4 — Evaluate the model

Test the model on examples not used during training. Compare predictions with known outcomes to see how well it works on unfamiliar data.

Step 5 — Make predictions

Give the trained model new input information. It uses learned patterns to produce a prediction, such as an estimated price for a new house.

Workflow: Collect Data → Features + Known Labels → Train → Evaluate → New Input → Prediction

3. Classification

Classification is a supervised learning task in which a model predicts a category or class.

Examples: an email is Spam or Not Spam; a student is predicted to Pass or Fail; a fruit is identified as Apple, Banana, Orange, or Mango.

Example: Email spam detection

Training examples contain email information and known labels such as Spam or Not Spam. The model learns patterns from these examples and predicts a class for a new email.

Memory trick: Classification predicts a category.

A. Binary classification

Binary classification predicts one of two possible classes.

- **Example:** Spam or Not Spam.
- **Example:** Pass or Fail.
- **Example:** Loan default or No loan default.

The model selects one of the two classes for the task.

B. Multiclass classification

Multiclass classification predicts one class from three or more possible classes. The model chooses one class for each prediction.

- **Fruit:** Apple, Banana, Orange, or Mango. For a banana, the model predicts Banana.
- **Animal:** Cat, Dog, Horse, or Elephant. If the task asks for the main animal, it predicts one class.
- **Student result:** Excellent, Good, Average, or Poor. The model predicts one category.

Remember: Multiclass = one class selected from many possible classes.

C. Multilabel classification

Multilabel classification allows one input to receive multiple labels at the same time.

- **Photo:** A photo may be labeled Dog, Person, Tree, and Car if those objects are present.
- **Movie:** One movie may receive Action, Comedy, and Adventure labels.
- **News:** One article may receive Politics, Economy, and Government labels.

Remember: Multiple labels can apply to one input. A particular input may still receive one label if only one label applies.

Multiclass vs. multilabel — the key difference

Suppose a photograph contains a dog and a cat.

- **Multiclass task:** If the model must identify the main animal and choose one class from Dog, Cat, Horse, or Elephant, it predicts one class.
- **Multilabel task:** If the model must identify all relevant objects, it can predict both Dog and Cat.

The difference is whether the task expects one class or allows multiple labels for each input, not simply how many categories exist.

4. Regression

Regression is a supervised learning task in which a model predicts a numerical value, often a quantity that can vary across a range.

Example: House-price prediction

A company has historical house records with features such as area, bedrooms, and location, together with actual selling prices. The model learns patterns from these labeled examples and predicts a price for a new house.

Other examples

- Food delivery: predict delivery time in minutes.
- Weather: predict temperature.
- Business: predict future sales.
- Housing: estimate a property's price.

Memory trick: Regression predicts a number; classification predicts a category.

Classification vs. regression

- Predict whether a student will Pass or Fail → Classification.
- Predict the student's exam score, such as 82 marks → Regression.
- Predict whether a house is expensive or inexpensive using defined categories → Classification.
- Predict the house's price in rupees → Regression.

5. What Happens When a Prediction Is Wrong?

A model can make incorrect predictions. During training, a **loss function** measures prediction error according to a chosen method. An **optimization** method adjusts the model's learned parameters to improve performance.

Simplified process: Make a prediction → Measure the error → Adjust the model → Repeat during training.

Training performance alone is not enough. A model should also be evaluated on data it did not train on, because good results on training examples do not guarantee good results on new examples.

6. Algorithm vs. Model

Algorithm: The method used to learn patterns from data.

Model: The learned system that uses those patterns to make predictions.

Learning algorithm + training data → trained model. The trained model can then receive new inputs and produce predictions.

7. Quick Revision

- **Supervised learning:** Learns from labeled training data.
- **Feature:** Input information used to make a prediction.
- **Label:** Known target answer in the training data.
- **Classification:** Predicts a category.
- **Binary classification:** One of two classes.
- **Multiclass classification:** One class from three or more possible classes.
- **Multilabel classification:** Multiple labels can be assigned to one input.
- **Regression:** Predicts a numerical value.
- **Loss function:** Measures prediction error using a chosen method.
- **Optimization:** Adjusts learned parameters to improve the training objective.
- **Evaluation:** Checks model performance on data not used for training.

8. Practice Task

A music app uses song information to make predictions. For one task, it predicts whether a song is Rock, Pop, Classical, or Hip-Hop. For another task, it predicts whether a song is energetic and whether it is romantic, allowing both labels to apply. Finally, it predicts how many minutes a song lasts.

Answer in plain sentences: Identify which task is multiclass classification, which is multilabel classification, and which is regression. Briefly explain why each task fits its category.

9. Key Takeaway

Supervised learning learns from labeled examples. Classification predicts categories, while regression predicts numerical values. Classification may be binary, multiclass, or multilabel depending on how the target classes and labels are defined.

End of Day 04 Notes